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Technology-Based Interventions for Nursing Home Residents: A Systematic Review Protocol

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Technology-Based Interventions for Nursing Home Residents: A Systematic Review Protocol

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Technology-Based Interventions for Nursing Home Residents: A Systematic Review Protocol

Abstract

Introduction: A growing number of technology-based interventions are used to support the health and quality of life of nursing home residents. The onset of COVID-19 and recommended social distancing policies that followed led to an increased interest in technology-based solutions to provide healthcare and promote health. Yet, there are no comprehensive resources on technology-based healthcare solutions that describe their efficacy for nursing home residents. This systematic review will identify technology-based interventions designed for nursing home residents and describe the characteristics and effects of these interventions concerning the distinctive traits of nursing home residents and nursing facilities. Additionally, this paper will present practical insights into the varying intervention approaches that can assist in the delivery of broad digital health solutions for nursing home residents amid and beyond the impact of COVID-19.

Methods and analysis: Databases including the PubMed, PsycINFO, CINAHL, and Scopus will be used to identify articles related to technology-based interventions for nursing home residents published between January 1st, 2010 to Sepetember 30th, 2021. Titles, abstracts, and full-text papers will be reviewed against the eligibility criteria. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses procedures will be followed for the reporting process, and implications for existing interventions and research evaluated by a multidisciplinary research team.

Ethics and dissemination: As the study is a protocol for a systematic review, ethical approval is not required. The study findings will be disseminated via peer-reviewed publications and conference presentations.

Systematic Review Registrations: PROSPERO CRD 42020191880

Keywords: COVID-19; novel coronavirus; SARS-CoV2; nursing home residents; older adults; underserved populations; technology-based interventions; socio-ecological model

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Strengths and limitations of this study

- This systematic review protocol follows the Preferred Reporting Items for Systematic Review and Meta- Analysis Protocols guidelines.
- To bridge the research gap, this systematic review will identify technology-based interventions designed for nursing home residents and describe the characteristics and effects of these interventions concerning the distinctive traits of nursing home residents and nursing facilities.
- Additionally, this paper will present practical insights into the varying intervention approaches that can assist in the delivery of broad digital health solutions for nursing home residents amid and beyond the impact of COVID-19.
- This review is limited to evidence within the scope of nursing home residents.
- Non-English databases will not be searched, which might limit the representativeness of the research findings.

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Technology-Based Interventions for Nursing Home Residents: A Systematic Review Protocol

Background

Nursing homes have been described as a "ground zero" throughout the coronavirus outbreak [1-4]. While the final impact of COVID-19 (coronavirus) on short and longer-term health outcomes is still unclear as the pandemic continues to unfold [5], what is clear is that nursing home residents have suffered some of the gravest consequences of this pandemic so far [6]. Contributing to over 40% of COVID-19 deaths within the United States, 100,033 residents and workers in nursing homes have passed away from COVID-19 (as of November 24th, 2020) [7]. Worse still is the fact that the nursing home residents who have succumbed to COVID-19 mostly died without the care or company of their family members [8-10]. These staggering numbers underscore the urgent need for healthcare researchers to understand factors that make nursing home residents more vulnerable to emergency events like COVID-19 [11], and to identify practical solutions that can address these factors in a timely fashion.

Nursing homes and nursing facilities provide long-term service and support for individuals living with chronic or disabling conditions who are unable to live at home independently [12-14]. Often living with multiple morbidities, in the United States approximately 85.1% of nursing home residents are 65 years and older, with 75.8% of these individuals experiencing hypertension, 58.9% living with Alzheimer's Disease, and 53.0% of residents living with depression [14]. Three sets of factors likely contribute to the alarming COVID-19 death rates seen in nursing homes: (1) the characteristics of nursing home residents, (2) the capabilities of nursing home facilities, and (3) the micro and macro-level supports available to nursing home residents [11]. On a micro level, research indicates that nursing home

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residents are more susceptible to infection and fatal outcomes from COVID-19 because they are often older adults living with medical conditions that compromise the immune systems and lowers their ability to combat the virus [14-17]. Additionally, they often lack specific self-care skills, such as utilizing telemedicine tools, or they may have a physical or cognitive impairment that impedes their ability to take care of their health and wellbeing [18-22]. The macro perspective focuses on the unique characteristics of nursing home facilities, as they are typically operating in a close and shared-living environment—conditions that are ideal for the spread of the virus [10, 17, 23]. Further, nursing homes often lack adequate healthcare resources or infrastructure needed to curb the impact of COVID-19. For example, numerous studies have indicated a lack of investment in training programs for nursing home staff, in addition to high turnover rates [24, 25], that management teams are often ineffective [26, 27], and that the nursing home infrastructure is often too outdated [23, 28-30].

The third set of factors that contribute to nursing home residents' vulnerability to COVID-19 centers on social support available to these adults [11]. Nursing home residents often have limited access to micro-level social support, including support from formal (e.g., doctors and nurses) or informal caregivers (e.g., family, friends, and acquaintances) [31], local community [32], and organizations (e.g., inexperienced or inadequately trained staff) [33]. Furthermore, nursing home residents often have limited macro-level social support. This is evidenced by harmful social norms (e.g., age-related discrimination)[34-36] and inadequate policy support that facilitates healthy aging and quality of life (e.g., insufficient regulatory oversight to ensure quality care in nursing homes) [4, 37-39]. These factors combined could result in severe health consequences in nursing home residents, such as wide viral spread [1].

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The above areas of inquiry resonate with the core principles of the socio-ecological model [40-43], which highlights the way individuals are influenced by a series of synergistic interactions between intrapersonal and interpersonal factors (e.g., residents, resident families), organizational characteristics (e.g., nursing homes), policy (e.g., legislative response) and the social/community (e.g., ageism) context, and how these processes can change over time (See Figure 1.) To successfully and effectively protect nursing home residents from global health crises like COVID-19, stakeholders such as policymakers, healthcare professionals, researchers informal caregivers, and older adults themselves all need to contribute to the change-making process (e.g., research safeguards or underscores the importance of COVID-19 protective measures [44-49]). While some effective changes are resource-intensive, time-consuming, and need concerted efforts from multilevel stakeholders to achieve, there are cost-effective, efficient, and accessible health solutions available to nursing home residents, such as technology-based interventions [50, 51].

---Insert Figure 1 here---

Technology-based interventions can be considered as the use of technology (e.g., digital devices like tablets and wearable devices, communication platforms) to manage or support health promotion strategies that aim to produce accessible and affordable health solutions to a target audience. Compared to traditional health solutions, such as face-to-face consultations, technology-based interventions have the potential to deliver healthcare more effectively and can mitigate geographic and access-related limitations that, as studies show, can play a significant role within nursing homes [52-58]. The evidence further suggests that technology-based

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interventions can help free healthcare professionals from repetitive work and allow them to make more meaningful contributions in delivering healthcare solutions to the care recipients [59-61].

Telemedicine and other technology-based solutions are particularly crucial given circumstances rendered by the COVID-19 pandemic, such as the limited ability for some healthcare providers to enter residences or for residents to visit their healthcare team for primary care visits. Limiting the exposure to infection through the use of telemedicine may assist in situations where a resident is required to attend a hospital appointment and return to a residence, thus alleviating the potential risk for a virus to spread to others [62]. Further, technologies that support residents' ability to remain in contact with families and friends outside of skilled care settings may reduce the adverse effects of loneliness and social isolation that are more common among nursing home residents compared to community-dwelling older adults [63, 64].

While technology-based solutions have the potential to deliver health solutions to nursing home residents [65, 66], there is limited awareness of the benefits and delivery options for state-of-art technology-based interventions specifically designed for nursing home residents. By factoring in the distinctive characteristics of nursing home residents and nursing home facilities, the main focus of this systematic review is to identify and evaluate technology-based interventions tailored explicitly for nursing home residents. Additionally, this research will present practical insights into the varying intervention approaches that can assist in the delivery of broad digital health solutions for nursing home residents amid and beyond the impact of COVID-19. Overall, our research aims are:

- To identify the characteristics and effects of technology-based interventions for nursing home residents.
- To identify effective technology-based interventions for nursing home residents that can be applied in the COVID-19 context.

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Methods

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) procedures will be adhered to in the reporting process [67]. This systematic review is registered with the International Prospective Register of Systematic Reviews (PROSPERO) system (CRD 42020191880); these measures are to avoid unnecessary study duplication [68, 69], increase research rigor [70, 71], improve study comparability and replicability [72], and ultimately, promote quality and transparency in research [73]. The review will be carried out in line with the Cochrane Handbook [74].

Inclusion and exclusion criteria

Based on the research aims, inclusion criteria were set a priori (Table 1). Articles published between January 1st, 2010 to September 30th, 2021 will be included in the review. This period of time was selected to ensure a focus on up-to-date technologies, given the tendency of technology-based interventions to evolve and become out-of-date. The timeline will also make sure both studies conducted before and amid COVID-19 will be included in the review. In this study, nursing home residents are defined as people "having a length of stay in a nursing home for more than 90 days at any point" [75]. Technology-based interventions are defined as "the use of technology to design, develop, and/or deliver health promotion contents and strategies that aim to induce or improve positive physical or psychological health outcomes" in nursing home residents [76]. In other words, interventions that: (1) incorporate technology (e.g., mobile applications), (2) utilize health promotional contents or strategies (e.g., information or emotional

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appeals), and (3) aim to improve nursing home residents' health outcomes (e.g., mental health status) will be included in the review.

Articles will be excluded if (1) the study sample did not include a majority of nursing home residents (i.e., nursing home residents ≤ 50% of the total research population), (2) the study did not focus on technology-based interventions (e.g., conventional interventions like face-to-face therapies), (3) study did not provide detailed information on the characteristics and effects of technology-based interventions (e.g., little or no information on the design or outcomes of the interventions), (4) the study did not report original and empirical findings on intervention outcomes (e.g., research protocols and systematic review studies), and (5) the study's main research method was not randomized controlled trials, experimental design, pre-post, quasi-experimental design, or observational studies.

---Insert Table 1 here---

Search strategy

Databases including PubMed, PsycINFO, CINAHL, and Scopus, will be searched for eligible articles. We will also search ProQuest Dissertations to examine gray literature sources. A search strategy was developed in consultation with a librarian experienced in systematic review methods. Search terms used to locate articles will center on three concepts: nursing home residents, technology-based interventions, and randomized controlled trials. An example PubMed search string is illustrated in Table 2.

---Insert Table 2 here---

Study selection

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Following the search, all citations will be collated and uploaded to Mendeley, and duplicate studies will be removed. Titles and then abstracts will be screened by two principal reviewers independently. The same screening procedure will be adopted in the full-text article review process on selected article abstracts. Reasons for exclusion will be recorded and detailed in the PRISMA flowchart. Discrepancies between reviewers will be resolved via group discussions using videoconferencing and email correspondence to reach a consensus.

Study quality assessment

The Cochrane Collaboration evaluation framework will be adopted to examine the risk of bias of the included study [77]. The framework has seven domains: random sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data, selective reporting, and any other source of bias. The risk of bias will be evaluated independently by two reviewers (ZS and XL), who will qualitatively evaluate the risk of bias and provide a score (high, medium, low). Any discrepancy regarding the risk of bias will be resolved by consensus via group discussions. Both junior and senior authors will be involved in the discrepancy review process, wherein, if agreement cannot be reached, the senior author will decide on the appropriate rating.

Data extraction and synthesis

Data on study design, sample characteristics (i.e., sample size and sample details), intervention characteristics (i.e., technology type, intervention application, intervention exposure, and intervention materials), outcome variables assessed, and research findings will be extracted by two main reviewers (ZS and XL). Findings from the included studies will be narratively

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synthesized to examine the characteristics and effects of the interventions. Gaining a more structured understanding of the interventions, the multidisciplinary study team will organize insights on intervention application and outcomes in tandem with the distinctive traits of nursing home residents and the overall nursing home environment. Due to the heterogeneity found within the articles identified during a preliminary review of search results, meta-analyses are not considered.

Ethics and dissemination:

As the study is a protocol for a systematic review, ethical approval is not required. The study findings will be disseminated via peer-reviewed publications and conference presentations.

Patient and Public Involvement

None.

Discussion

There is a growing body of technology-based interventions designed to support the health and quality of life of nursing home residents [52-58]. The onset of COVID-19 and recommended social distancing policy led to an increased interest in reliance on technology-based solutions [78, 79]. However, research has yet to provide comparative insight into the recent state of development of these interventions and how current evidence can be applied to the context of COVID-19. The use of the socio-ecological model, combined with multidisciplinary expertise,

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provides a framework to present practical insights on how these interventions can be utilized to deliver health solutions to nursing home residents amid and beyond the impact of COVID-19.

This research fills a critical gap in the literature by consolidating, in one place, the evidence for technology-based interventions empirically tested with nursing home populations. As the older adult population grows, there is an urgent need to identify effective technology-based interventions that can address the distinctive characteristics and preferences of nursing home residents [80, 81]. Improving person-centered care and the delivery of effective care solutions to nursing home residents, especially as the pandemic continues, is of critical importance. A comprehensive understanding of how available technology-based health solutions facilitate healthcare for nursing home residents can help shed light on approaches that are available to these residents to fend off the negative health consequences amid and beyond the influence of COVID-19. While the COVID-19 pandemic has revealed troubling vulnerabilities in the long-term care system across the globe, it also shows how telemedicine can support nursing home residents and their families. Technology can also assist clinicians in connecting with patients when in-person medical visits are difficult or dangerous (e.g., in rural settings, following natural disasters). Telemedicine and other technology-based interventions have the potential to provide a comprehensive range of benefits. This research also serves as a platform for care institutions and policymakers to inform government policies and further justify the role technology can play in strengthening the service provision across nursing homes and facilities.

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List of abbreviation

COVID-19: covronavirus disease 2019

Declarations

- Ethics approval and consent to participate
  - Not applicable.
- Consent for publication
  - Not applicable.
- Availability of data and materials
  - Data are available upon reasonable request.
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Table 1. Study inclusion criteria

Category	Criteria
Study population	Adults (≥18 years old) living in nursing homes
Language	English
Intervention/health solution	Technology-based interventions (e.g., digital tools such as smartphones and tablets, sensor devices, internet-based programs)
Key variable	Detailed descriptions of the technology-based interventions (i.e., purpose of the intervention (to evaluate the aims of existing interventions), use of technology (to examine how different types of technologies have been applied among nursing home residents), application of the interventions (to investigate the degree to which the interventions have involved nursing home residents in the adoption/application of technologies), intervention exposure (to study how different interventions have been used among nursing home residents), outcome variables assessed/measured (to evaluate the outcomes of technology-based interventions, such as improved physical or psychological health or wellbeing; more satisfactory patient-provider communication), and weather the design of the intervention material is tailored to nursing home residents (to examine to degree to which the design technology-based interventions has taken the unique attributes of nursing home residents into consideration)
Study type	Empirical studies
Study design	Randomized controlled trials, experimental design, pre-post, quasi-experimental design, and observational studies
Study outcome	Empirical reporting of the effect of the intervention (i.e., qualitative designs excluded)

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Table 2. Example PubMed search string

Concept	Search string
Nursing homes	"nursing home"[MeSH] OR "nursing home"[TIAB] OR "nursing homes"[MeSH] OR "nursing homes"[TIAB] OR "residential home*"[MeSH] OR "residential home"[TIAB] OR "caring home*"[TIAB] OR "home for the aged" [MeSH] "home for the aged" [TIAB] OR "long term care"[MeSH] OR "long term care"[TIAB] OR "senior housing"[TIAB] OR "assisted living facilities"[MeSH] OR "assisted living facilities"[TIAB] OR (senior[TIAB] OR geriatric[TIAB] OR elderly[TIAB] OR aged[TIAB] OR elder[TIAB] "older adults"[TIAB]) AND (housing[TIAB] OR living[TIAB])
Technology-based Interventions	"technology"[MeSH] OR "technology"[TIAB] OR "eHealth"[TIAB] OR "telemedicine"[MeSH] OR "telemedicine"[TIAB] OR "tele-medicine"[MeSH] OR "tele-medicine"[TIAB] OR "telehealth"[TIAB] OR "tele-health"[TIAB] OR "connected health"[TIAB] OR "digital health"[TIAB] OR "mHealth"[TIAB] OR "mobile health"[TIAB]
Randomized controlled trials	randomized controlled trial[PT] OR randomized controlled trials as topic[MH] OR random allocation [MH] OR double-blind method[MH] OR single-blind method[MH] OR random*[tw] OR "Placebos"[MeSH] OR placebo[TIAB] OR ((singl*[tw] OR doubl*[tw] OR trebl*[TW] OR tripl*[TW]) AND (mask*[TW] OR blind*[TW] OR dumm*[TW]))
COVID-19	((coronavirus OR "corona virus" OR coronavirinae OR coronaviridae OR betacoronavirus OR covid19 OR "covid 19" OR nCoV OR "CoV 2" OR CoV2 OR sarscov2 OR 2019nCoV OR "novel CoV" OR "wuhan virus") OR ((wuhan OR hubei OR huanan) AND ("severe acute respiratory" OR pneumonia) AND (outbreak)) OR "Coronavirus"[Mesh] OR "Coronavirus Infections"[Mesh] OR "COVID-19" [Supplementary Concept] OR "severe acute respiratory syndrome coronavirus 2" [Supplementary Concept] OR "Betacoronavirus"[Mesh])

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Figure 1. An ecological model of factors increasing nursing home resident vulnerability to COVID-19

Please refer to the supplementary materials section for the figure

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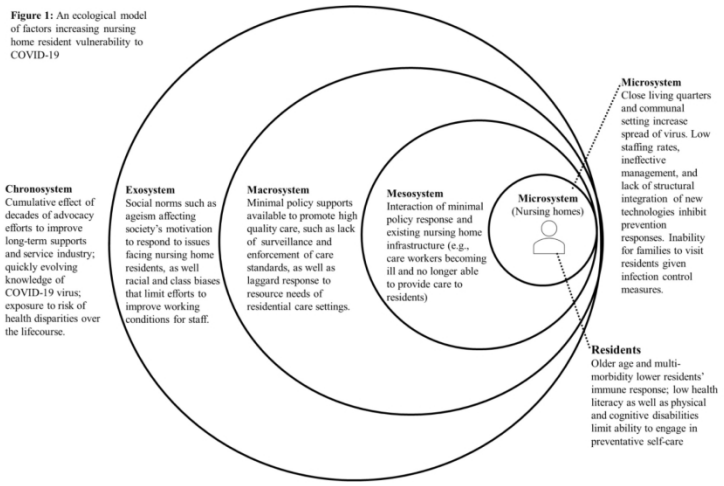


Figure 1. An ecological model of factors increasing nursing home resident vulnerability to COVID-19

899x1164mm (72 x 72 DPI)



PRISMA Checklist

Section and Topic	Item #	Checklist item	Reported
TITLE			
Title	1	Identify the report as a systematic review.	√
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	√
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	√
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	√
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	√
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	√
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	√
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	√
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	√
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	√
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	√
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	√
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	√
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	√
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	√
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	√
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	√
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	√
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	√
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	√

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PRISMA Checklist

Section and Topic	Item #	Checklist item	Reported
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	√
Study characteristics	17	Cite each included study and present its characteristics.	√
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	√
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	√
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	√
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	√
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	√
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	√
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	√
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	√
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	√
	23b	Discuss any limitations of the evidence included in the review.	√
	23c	Discuss any limitations of the review processes used.	√
	23d	Discuss implications of the results for practice, policy, and future research.	√
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	√
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	√
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	√
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	√
Competing interests	26	Declare any competing interests of review authors.	√
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	√

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Technology-Based Interventions for Nursing Home Residents: A Systematic Review Protocol

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Technology-Based Interventions for Nursing Home Residents: A Systematic Review Protocol

Abstract

Introduction: A growing number of technology-based interventions are used to support the health and quality of life of nursing home residents. The onset of COVID-19 and recommended social distancing policies that followed led to an increased interest in technology-based solutions to provide healthcare and promote health. Yet, there are no comprehensive resources on technology-based healthcare solutions that describe their efficacy for nursing home residents. This systematic review will identify technology-based interventions designed for nursing home residents and describe the characteristics and effects of these interventions concerning the distinctive traits of nursing home residents and nursing facilities. Additionally, this paper will present practical insights into the varying intervention approaches that can assist in the delivery of broad digital health solutions for nursing home residents amid and beyond the impact of COVID-19.

Methods and analysis: Databases including the PubMed, PsycINFO, CINAHL, and Scopus will be used to identify articles related to technology-based interventions for nursing home residents published between January 1st, 2010 to September 30th, 2021. Titles, abstracts, and full-text papers will be reviewed against the eligibility criteria. The Cochrane Collaboration evaluation framework will be adopted to examine the risk of bias of the included study. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses procedures will be followed for the reporting process, and implications for existing interventions and research evaluated by a multidisciplinary research team.

Ethics and dissemination: As the study is a protocol for a systematic review, ethical approval is not required. The study findings will be disseminated via peer-reviewed publications and conference presentations.

Systematic Review Registrations: PROSPERO CRD 42020191880

Keywords: COVID-19; novel coronavirus; SARS-CoV2; nursing home residents; older adults; underserved populations; technology-based interventions; socio-ecological model

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Strengths and limitations of this study

- This systematic review protocol follows the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols guidelines.
- The review will systematically evaluate the impact of technology-based interventions for nursing home residents, such as the characteristics and effects of these interventions on the health outcomes of these residents amid the COVID-19 pandemic.
- Our paper will also examine the delivery mode of the interventions (e.g., mobile-based vs. computer-based) for nursing home residents utilized amid the COVID-19 pandemic.
- This review is limited to evidence within the scope of nursing home residents and technology-based interventions.
- Non-English databases will not be searched, which might limit the representativeness of the research findings.

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Technology-Based Interventions for Nursing Home Residents: A Systematic Review Protocol

Background

Nursing homes have been described as a "ground zero" throughout the coronavirus outbreak [1-4]. While the final impact of COVID-19 (coronavirus) on short and longer-term health outcomes is still unclear as the pandemic continues to unfold [5], what is clear is that nursing home residents have suffered some of the gravest consequences of this pandemic so far [6]. Contributing to over 40% of COVID-19 deaths within the United States, 100,033 residents and workers in nursing homes have passed away from COVID-19 (as of November 24th, 2020) [7]. Worse still is the fact that the nursing home residents who have succumbed to COVID-19 mostly died without the care or company of their family members [8-10]. These staggering numbers underscore the urgent need for healthcare researchers to understand factors that make nursing home residents more vulnerable to emergency events like COVID-19 [11], and to identify practical solutions that can address these factors in a timely fashion.

Nursing homes and nursing facilities provide long-term service and support for individuals living with chronic or disabling conditions who are unable to live at home independently [12-14]. Often living with multiple morbidities, in the United States approximately 85.1% of nursing home residents are 65 years and older, with 75.8% of these individuals experiencing hypertension, 58.9% living with Alzheimer's Disease, and 53.0% of residents living with depression [14]. Three sets of factors likely contribute to the alarming COVID-19 death rates seen in nursing homes: (1) the characteristics of nursing home residents, (2) the capabilities of nursing home facilities, and (3) the micro and macro-level supports available to nursing home residents [11]. On a micro level, research indicates that nursing home

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residents are more susceptible to infection and fatal outcomes from COVID-19 because they are often older adults living with medical conditions that compromise the immune systems and lowers their ability to combat the virus [14-17]. Additionally, they often lack specific self-care skills, such as utilizing telemedicine tools, or they may have a physical or cognitive impairment that impedes their ability to take care of their health and wellbeing [18-22]. The macro perspective focuses on the unique characteristics of nursing home facilities, as they are typically operating in a close and shared-living environment—conditions that are ideal for the spread of the virus [10, 17, 23]. Further, nursing homes often lack adequate healthcare resources or infrastructure needed to curb the impact of COVID-19. For example, numerous studies have indicated a lack of investment in training programs for nursing home staff, in addition to high turnover rates [24, 25], that management teams are often ineffective [26, 27], and that the nursing home infrastructure is often too outdated [23, 28-30].

The third set of factors that contribute to nursing home residents' vulnerability to COVID-19 centers on social support available to these adults [11]. Nursing home residents often have limited access to micro-level social support, including support from formal (e.g., doctors and nurses) or informal caregivers (e.g., family, friends, and acquaintances) [31], local community [32], and organizations (e.g., inexperienced or inadequately trained staff) [33]. Furthermore, nursing home residents often have limited macro-level social support. This is evidenced by harmful social norms (e.g., age-related discrimination)[34-36] and inadequate policy support that facilitates healthy aging and quality of life (e.g., insufficient regulatory oversight to ensure quality care in nursing homes) [4, 37-39]. These factors combined could result in severe health consequences in nursing home residents, such as wide viral spread [1].

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The above areas of inquiry resonate with the core principles of the socio-ecological model [40-43], which highlights the way individuals are influenced by a series of synergistic interactions between intrapersonal and interpersonal factors (e.g., residents, resident families), organizational characteristics (e.g., nursing homes), policy (e.g., legislative response) and the social/community (e.g., ageism) context, and how these processes can change over time (See Figure 1.) To successfully and effectively protect nursing home residents from global health crises like COVID-19, stakeholders such as policymakers, healthcare professionals, researchers informal caregivers, and older adults themselves all need to contribute to the change-making process (e.g., research safeguards or underscores the importance of COVID-19 protective measures [44-49]). While some effective changes are resource-intensive, time-consuming, and need concerted efforts from multilevel stakeholders to achieve, there are cost-effective, efficient, and accessible health solutions available to nursing home residents, such as technology-based interventions [50, 51].

---Insert Figure 1 here---

Technology-based interventions can be considered as the use of technology (e.g., digital devices like tablets and wearable devices, communication platforms) to manage or support health promotion strategies that aim to produce accessible and affordable health solutions to a target audience. Compared to traditional health solutions, such as face-to-face consultations, technology-based interventions have the potential to deliver healthcare more effectively and can mitigate geographic and access-related limitations that, as studies show, can play a significant role within nursing homes [52-58]. The evidence further suggests that technology-based

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interventions can help free healthcare professionals from repetitive work and allow them to make more meaningful contributions in delivering healthcare solutions to the care recipients [59-61].

Telemedicine and other technology-based solutions are particularly crucial given circumstances rendered by the COVID-19 pandemic, such as the limited ability for some healthcare providers to enter residences or for residents to visit their healthcare team for primary care visits. Limiting the exposure to infection through the use of telemedicine may assist in situations where a resident is required to attend a hospital appointment and return to a residence, thus alleviating the potential risk for a virus to spread to others [62]. Further, technologies that support residents' ability to remain in contact with families and friends outside of skilled care settings may reduce the adverse effects of loneliness and social isolation that are more common among nursing home residents compared to community-dwelling older adults [63, 64].

While technology-based solutions have the potential to deliver health solutions to nursing home residents [65, 66], there is limited awareness of the benefits and delivery options for state-of-art technology-based interventions specifically designed for nursing home residents. By factoring in the distinctive characteristics of nursing home residents and nursing home facilities, the main focus of this systematic review is to identify and evaluate technology-based interventions tailored explicitly for nursing home residents. Additionally, our study will examine the delivery mode of the interventions, such as mobile-based vs. computer-based interventions and interventions that require nursing home residents' input/engagement vs. those that do not require users' input/engagement (e.g., sensor-based interventions) amid the COVID-19 pandemic.

Overall, our research aims are:

- To identify the characteristics and effects of technology-based interventions for nursing home residents.

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- To identify effective technology-based interventions for nursing home residents that can be applied in the COVID-19 context.

Methods

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) procedures will be adhered to in the reporting process (please see the supplementary file) [67]. This systematic review is registered with the International Prospective Register of Systematic Reviews (PROSPERO) system (CRD 42020191880); these measures are to avoid unnecessary study duplication [68, 69], increase research rigor [70, 71], improve study comparability and replicability [72], and ultimately, promote quality and transparency in research [73]. The review will be carried out in line with the Cochrane Handbook [74].

Inclusion and exclusion criteria

Based on the research aims, inclusion criteria were set a priori (Table 1). Articles published between January 1st, 2010 to September 30th, 2021 will be included in the review. This time period was selected to ensure a focus on up-to-date technologies, given the tendency of technology-based interventions to evolve and become out-of-date. The timeline will also make sure both studies conducted before and amid COVID-19 will be included in the review. In this study, nursing home residents are defined as people "having a length of stay in a nursing home for more than 90 days at any point" [75]. Technology-based interventions are defined as "the use of technology to design, develop, and/or deliver health promotion contents and strategies that aim to induce or improve positive physical or psychological health outcomes" in nursing home residents [76]. In other words, interventions that: (1) incorporate technology (e.g., mobile

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applications), (2) utilize health promotional contents or strategies (e.g., information or emotional appeals), and (3) aim to improve nursing home residents' health outcomes (e.g., mental health status) will be included in the review.

It is important to note that, in this study, technology-based interventions are considered conceptually different from phenomena such as digital solutions. Compared to concepts such as digital solutions, technology-based interventions are unique as they are: (1) evidence-based: one of the main objectives of this study is to investigate and identify interventions that could offer the most benefits to nursing home residents based on empirical evidence, (2) purposeful: technology-based interventions are designed, developed, and/or delivered often tailored to the unique attributes of the target audience (e.g., nursing home residents), and (3) broader in scale and scope: technology-based interventions include interventions that are developed based on non-digital technologies (those that based on systems that generate and process binary data like 0 and 1), like landline telephones and audiotapes, which adopts analog technology (stores and interprets information in line with the devices physical properties) [77].

Overall, articles will be excluded if (1) the study sample did not include a majority of nursing home residents (i.e., nursing home residents ≤ 50% of the total research population), (2) the study did not focus on technology-based interventions (e.g., conventional interventions like face-to-face therapies), (3) study did not provide detailed information on the characteristics and effects of technology-based interventions (e.g., little or no information on the design or outcomes of the interventions), (4) the study did not report original and empirical findings on intervention outcomes (e.g., research protocols and systematic review studies), and (5) the study's main research method was not randomized controlled trials, experimental design, pre-post, quasi-experimental design, or observational studies.

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---Insert Table 1 here---

Search strategy

Databases including PubMed, PsycINFO, CINAHL, and Scopus, will be searched for eligible articles. We will also search ProQuest Dissertations to examine gray literature sources. A search strategy was developed in consultation with a librarian experienced in systematic review methods. Search terms used to locate articles will center on three concepts: nursing home residents, technology-based interventions, and randomized controlled trials. An example PubMed search string is illustrated in Table 2. Similar search terms will be adopted that are in accordance with: (1) the scale and scope of the databases and (2) insights and evidence accumulated in the review process.

---Insert Table 2 here---

Study selection

Following the search, all citations will be collated and uploaded to Mendeley, and duplicate studies will be removed. Titles and then abstracts will be screened by two principal reviewers independently. The same screening procedure will be adopted in the full-text article review process on selected article abstracts. Reasons for exclusion will be recorded and detailed in the PRISMA flowchart. Discrepancies between reviewers will be resolved via group discussions using videoconferencing and email correspondence to reach a consensus.

Study quality assessment

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The Cochrane Collaboration evaluation framework will be adopted to examine the risk of bias of the included study [78]. The framework has seven domains: random sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data, selective reporting, and any other source of bias. The risk of bias will be evaluated independently by two reviewers (ZS and XL), who will qualitatively evaluate the risk of bias and provide a score (high, medium, low). Any discrepancy regarding the risk of bias will be resolved by consensus via group discussions. Both junior and senior authors will be involved in the discrepancy review process, wherein, if agreement cannot be reached, the senior author will decide on the appropriate rating.

Data extraction and synthesis

Data on study design, sample characteristics (i.e., sample size and sample details), intervention characteristics (i.e., technology type, intervention application, intervention exposure, and intervention materials), outcome variables assessed, and research findings will be extracted by two main reviewers (ZS and XL). Findings from the included studies will be narratively synthesized to examine the characteristics and effects of the interventions. Gaining a more structured understanding of the interventions, the multidisciplinary study team will organize insights on intervention application and outcomes in tandem with the distinctive traits of nursing home residents and the overall nursing home environment. Due to the heterogeneity found within the articles identified during a preliminary review of search results, meta-analyses are not considered.

Ethics and dissemination:

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As the study is a protocol for a systematic review, ethical approval is not required. Ethical research practices, such as the guideline proposed by the United States National Institutes of Health [79], will be observed and abided by throughout the research continuum. The study findings will be disseminated via peer-reviewed publications and conference presentations.

Patient and Public Involvement

None.

Discussion

There is a growing body of technology-based interventions designed to support the health and quality of life of nursing home residents [52-58]. The onset of COVID-19 and recommended social distancing policy led to an increased interest in reliance on technology-based solutions [80, 81]. However, research has yet to provide comparative insight into the recent state of development of these interventions and how current evidence can be applied to the context of COVID-19. The use of the socio-ecological model, combined with multidisciplinary expertise, provides a framework to present practical insights on how these interventions can be utilized to deliver health solutions to nursing home residents amid and beyond the impact of COVID-19.

This research fills a critical gap in the literature by consolidating, in one place, the evidence for technology-based interventions empirically tested with nursing home populations. As the older adult population grows, there is an urgent need to identify effective technology-based interventions that can address the distinctive characteristics and preferences of nursing home residents [82, 83]. Improving person-centered care and the delivery of effective care solutions to nursing home residents, especially as the pandemic continues, is of critical

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importance. A comprehensive understanding of how available technology-based health solutions facilitate healthcare for nursing home residents can help shed light on approaches that are available to these residents to fend off the negative health consequences amid and beyond the influence of COVID-19. While the COVID-19 pandemic has revealed troubling vulnerabilities in the long-term care system across the globe, it also shows how telemedicine can support nursing home residents and their families. Technology can also assist clinicians in connecting with patients when in-person medical visits are difficult or dangerous (e.g., in rural settings, following natural disasters). Telemedicine and other technology-based interventions have the potential to provide a comprehensive range of benefits. This research also serves as a platform for care institutions and policymakers to inform government policies and further justify the role technology can play in strengthening the service provision across nursing homes and facilities.

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List of abbreviation

COVID-19: covronavirus disease 2019

Declarations

- Ethics approval and consent to participate
  - Not applicable.
- Consent for publication
  - Not applicable.
- Availability of data and materials
  - Data are available upon reasonable request.
- Competing interests
  - None.
- Funding
  - None.
- Authors' contributions
  - ZS developed the research idea and wrote and manuscript. KM, YL, DMD, NMJ, XL, YD, SA, AC, JA, CPV, RYNC, JW, and HD reviewed and revised the manuscript.
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Table 1. Study inclusion criteria

Category	Criteria
Study population	Adults (≥18 years old) living in nursing homes
Language	English
Intervention/health solution	Technology-based interventions (e.g., educational materials delivered via smartphones and tablets, sensor devices, internet-based programs, etc.)
Key variable	Detailed descriptions of the technology-based interventions (i.e., purpose of the intervention (to evaluate the aims of existing interventions), use of technology (to examine how different types of technologies have been applied among nursing home residents), application of the interventions (to investigate the degree to which the interventions have involved nursing home residents in the adoption/application of technologies), intervention exposure (to study how different interventions have been used among nursing home residents), outcome variables assessed/measured (to evaluate the outcomes of technology-based interventions, such as improved physical or psychological health or wellbeing; more satisfactory patient-provider communication), and weather the design of the intervention material is tailored to nursing home residents (to examine to degree to which the design technology-based interventions has taken the unique attributes of nursing home residents into consideration)
Study type	Empirical studies
Study design	Randomized controlled trials, experimental design, pre-post, quasi-experimental design, and observational studies
Study outcome	Empirical reporting of the effect of the intervention (i.e., qualitative designs excluded)

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Table 2. Example PubMed search string

Concept	Search string
Nursing homes	"nursing home"[MeSH] OR "nursing home"[TIAB] OR "nursing homes"[MeSH] OR "nursing homes"[TIAB] OR "residential home*"[MeSH] OR "residential home"[TIAB] OR "caring home*"[TIAB] OR "home for the aged" [MeSH] "home for the aged" [TIAB] OR "long term care"[MeSH] OR "long term care"[TIAB] OR "senior housing"[TIAB] OR "assisted living facilities"[MeSH] OR "assisted living facilities"[TIAB] OR (senior[TIAB] OR geriatric[TIAB] OR elderly[TIAB] OR aged[TIAB] OR elder[TIAB] "older adults"[TIAB]) AND (housing[TIAB] OR living[TIAB])
Technology-based Interventions	"technology"[MeSH] OR "technology"[TIAB] OR "eHealth"[TIAB] OR "telemedicine"[MeSH] OR "telemedicine"[TIAB] OR "tele-medicine"[MeSH] OR "tele-medicine"[TIAB] OR "telehealth"[TIAB] OR "tele-health"[TIAB] OR "connected health"[TIAB] OR "digital health"[TIAB] OR "mHealth"[TIAB] OR "mobile health"[TIAB]
Randomized controlled trials	randomized controlled trial[PT] OR randomized controlled trials as topic[MH] OR random allocation [MH] OR double-blind method[MH] OR single-blind method[MH] OR random*[tw] OR "Placebos"[MeSH] OR placebo[TIAB] OR ((singl*[tw] OR doubl*[tw] OR trebl*[TW] OR tripl*[TW]) AND (mask*[TW] OR blind*[TW] OR dumm*[TW]))
COVID-19	((coronavirus OR "corona virus" OR coronavirinae OR coronaviridae OR betacoronavirus OR covid19 OR "covid 19" OR nCoV OR "CoV 2" OR CoV2 OR sarscov2 OR 2019nCoV OR "novel CoV" OR "wuhan virus") OR ((wuhan OR hubei OR huanan) AND ("severe acute respiratory" OR pneumonia) AND (outbreak)) OR "Coronavirus"[Mesh] OR "Coronavirus Infections"[Mesh] OR "COVID-19" [Supplementary Concept] OR "severe acute respiratory syndrome coronavirus 2" [Supplementary Concept] OR "Betacoronavirus"[Mesh])

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Figure 1. An ecological model of factors increasing nursing home resident vulnerability to COVID-19

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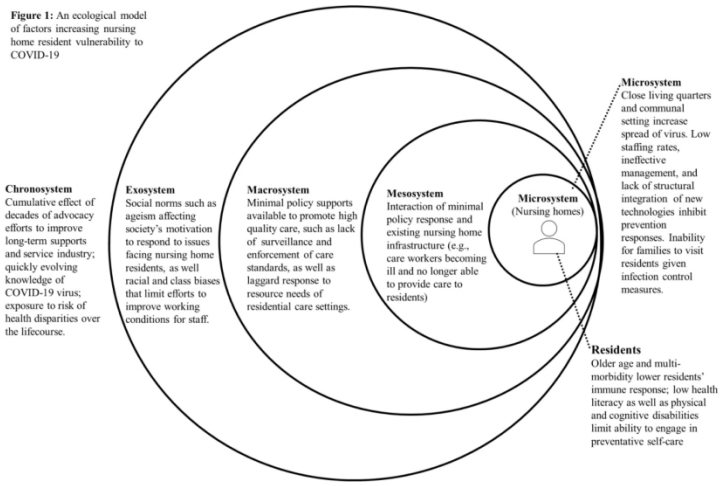


Figure 1. An ecological model of factors increasing nursing home resident vulnerability to COVID-19

899x1164mm (72 x 72 DPI)

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Page
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	2
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	NA
Support:			
Sources	5a	Indicate sources of financial or other support for the review	14
Sponsor	5b	Provide name for the review funder and/or sponsor	
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	4-7
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4-7
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	8-11
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	8-9
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	10
Study records:			

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Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	10-11
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	10-11
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	10-11
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	10-11
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	20
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	10-11
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	11
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	11
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	10-11
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	10-11
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	11
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	10-11

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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Technology-Based Interventions for Nursing Home Residents: A Systematic Review Protocol

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Technology-Based Interventions for Nursing Home Residents: A Systematic Review Protocol

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Technology-Based Interventions for Nursing Home Residents: A Systematic Review Protocol

Abstract

Introduction: A growing number of technology-based interventions are used to support the health and quality of life of nursing home residents. The onset of COVID-19 and recommended social distancing policies that followed led to an increased interest in technology-based solutions to provide healthcare and promote health. Yet, there are no comprehensive resources on technology-based healthcare solutions that describe their efficacy for nursing home residents. This systematic review will identify technology-based interventions designed for nursing home residents and describe the characteristics and effects of these interventions concerning the distinctive traits of nursing home residents and nursing facilities. Additionally, this paper will present practical insights into the varying intervention approaches that can assist in the delivery of broad digital health solutions for nursing home residents amid and beyond the impact of COVID-19.

Methods and analysis: Databases including the PubMed, PsycINFO, CINAHL, and Scopus will be used to identify articles related to technology-based interventions for nursing home residents published between January 1st, 2010 to September 30th, 2021. Titles, abstracts, and full-text papers will be reviewed against the eligibility criteria. The Cochrane Collaboration evaluation framework will be adopted to examine the risk of bias of the included study. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses procedures will be followed for the reporting process, and implications for existing interventions and research evaluated by a multidisciplinary research team.

Ethics and dissemination: As the study is a protocol for a systematic review, ethical approval is not required. The study findings will be disseminated via peer-reviewed publications and conference presentations.

Systematic Review Registrations: PROSPERO CRD 42020191880

Keywords: COVID-19; novel coronavirus; SARS-CoV2; nursing home residents; older adults; underserved populations; technology-based interventions; socio-ecological model

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Strengths and limitations of this study

- This systematic review protocol follows the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols guidelines.
- Our review will systematically evaluate the impact of technology-based interventions for nursing home residents.
- The social ecological model will be adopted to theoretically and systematically guide our review process.
- The current study is limited to evidence within the scope of nursing home residents and technology-based interventions.
- Non-English databases will not be searched, which might limit the representativeness of the research findings.

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Technology-Based Interventions for Nursing Home Residents: A Systematic Review Protocol

Background

Nursing homes have been described as a "ground zero" throughout the coronavirus outbreak [1-4]. While the final impact of COVID-19 (coronavirus) on short and longer-term health outcomes is still unclear as the pandemic continues to unfold [5], what is clear is that nursing home residents have suffered some of the gravest consequences of this pandemic so far [6]. Contributing to over 40% of COVID-19 deaths within the United States, 100,033 residents and workers in nursing homes have passed away from COVID-19 (as of November 24th, 2020) [7]. Worse still is the fact that the nursing home residents who have succumbed to COVID-19 mostly died without the care or company of their family members [8-10]. These staggering numbers underscore the urgent need for healthcare researchers to understand factors that make nursing home residents more vulnerable to emergency events like COVID-19 [11], and to identify practical solutions that can address these factors in a timely fashion.

Nursing homes and nursing facilities provide long-term service and support for individuals living with chronic or disabling conditions who are unable to live at home independently [12-14]. Often living with multiple morbidities, in the United States approximately 85.1% of nursing home residents are 65 years and older, with 75.8% of these individuals experiencing hypertension, 58.9% living with Alzheimer's Disease, and 53.0% of residents living with depression [14]. Three sets of factors likely contribute to the alarming COVID-19 death rates seen in nursing homes: (1) the characteristics of nursing home residents, (2) the capabilities of nursing home facilities, and (3) the micro and macro-level supports available to nursing home residents [11]. On a micro level, research indicates that nursing home

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residents are more susceptible to infection and fatal outcomes from COVID-19 because they are often older adults living with medical conditions that compromise the immune systems and lowers their ability to combat the virus [14-17]. Additionally, they often lack specific self-care skills, such as utilizing telemedicine tools, or they may have a physical or cognitive impairment that impedes their ability to take care of their health and wellbeing [18-22]. The macro perspective focuses on the unique characteristics of nursing home facilities, as they are typically operating in a close and shared-living environment—conditions that are ideal for the spread of the virus [10, 17, 23]. Further, nursing homes often lack adequate healthcare resources or infrastructure needed to curb the impact of COVID-19. For example, numerous studies have indicated a lack of investment in training programs for nursing home staff, in addition to high turnover rates [24, 25], that management teams are often ineffective [26, 27], and that the nursing home infrastructure is often too outdated [23, 28-30].

The third set of factors that contribute to nursing home residents' vulnerability to COVID-19 centers on social support available to these adults [11]. Nursing home residents often have limited access to micro-level social support, including support from formal (e.g., doctors and nurses) or informal caregivers (e.g., family, friends, and acquaintances) [31], local community [32], and organizations (e.g., inexperienced or inadequately trained staff) [33]. Furthermore, nursing home residents often have limited macro-level social support. This is evidenced by harmful social norms (e.g., age-related discrimination)[34-36] and inadequate policy support that facilitates healthy aging and quality of life (e.g., insufficient regulatory oversight to ensure quality care in nursing homes) [4, 37-39]. These factors combined could result in severe health consequences in nursing home residents, such as wide viral spread [1].

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The above areas of inquiry resonate with the core principles of the socio-ecological model [40-43], which highlights the way individuals are influenced by a series of synergistic interactions between intrapersonal and interpersonal factors (e.g., residents, resident families), organizational characteristics (e.g., nursing homes), policy (e.g., legislative response) and the social/community (e.g., ageism) context, and how these processes can change over time (See Figure 1.) To successfully and effectively protect nursing home residents from global health crises like COVID-19, stakeholders such as policymakers, healthcare professionals, researchers informal caregivers, and older adults themselves all need to contribute to the change-making process (e.g., research safeguards or underscores the importance of COVID-19 protective measures [44-49]). While some effective changes are resource-intensive, time-consuming, and need concerted efforts from multilevel stakeholders to achieve, there are cost-effective, efficient, and accessible health solutions available to nursing home residents, such as technology-based interventions [50, 51].

---Insert Figure 1 here---

Technology-based interventions can be considered as the use of technology (e.g., digital devices like tablets and wearable devices, communication platforms) to manage or support health promotion strategies that aim to produce accessible and affordable health solutions to a target audience. Compared to traditional health solutions, such as face-to-face consultations, technology-based interventions have the potential to deliver healthcare more effectively and can mitigate geographic and access-related limitations that, as studies show, can play a significant role within nursing homes [52-58]. The evidence further suggests that technology-based

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interventions can help free healthcare professionals from repetitive work and allow them to make more meaningful contributions in delivering healthcare solutions to the care recipients [59-61].

Telemedicine and other technology-based solutions are particularly crucial given circumstances rendered by the COVID-19 pandemic, such as the limited ability for some healthcare providers to enter residences or for residents to visit their healthcare team for primary care visits. Limiting the exposure to infection through the use of telemedicine may assist in situations where a resident is required to attend a hospital appointment and return to a residence, thus alleviating the potential risk for a virus to spread to others [62]. Further, technologies that support residents' ability to remain in contact with families and friends outside of skilled care settings may reduce the adverse effects of loneliness and social isolation that are more common among nursing home residents compared to community-dwelling older adults [63, 64].

While technology-based solutions have the potential to deliver health solutions to nursing home residents [65, 66], there is limited awareness of the benefits and delivery options for state-of-art technology-based interventions specifically designed for nursing home residents. By factoring in the distinctive characteristics of nursing home residents and nursing home facilities, the main focus of this systematic review is to identify and evaluate technology-based interventions tailored explicitly for nursing home residents. Additionally, our study will examine the delivery mode of the interventions, such as mobile-based vs. computer-based interventions and interventions that require nursing home residents' input/engagement vs. those that do not require users' input/engagement (e.g., sensor-based interventions) amid the COVID-19 pandemic.

Overall, our research aims are:

- To identify the characteristics and effects of technology-based interventions for nursing home residents.

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- To identify effective technology-based interventions for nursing home residents that can be applied in the COVID-19 context.

Methods

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) procedures will be adhered to in the reporting process (please see the supplementary file) [67]. This systematic review is registered with the International Prospective Register of Systematic Reviews (PROSPERO) system (CRD 42020191880); these measures are to avoid unnecessary study duplication [68, 69], increase research rigor [70, 71], improve study comparability and replicability [72], and ultimately, promote quality and transparency in research [73]. The review will be carried out in line with the Cochrane Handbook [74].

Inclusion and exclusion criteria

Based on the research aims, inclusion criteria were set a priori (Table 1). Articles published between January 1st, 2010 to September 30th, 2021 will be included in the review. This time period was selected to ensure a focus on up-to-date technologies, given the tendency of technology-based interventions to evolve and become out-of-date. The timeline will also make sure both studies conducted before and amid COVID-19 will be included in the review. In this study, nursing home residents are defined as people "having a length of stay in a nursing home for more than 90 days at any point" [75]. Technology-based interventions are defined as "the use of technology to design, develop, and/or deliver health promotion contents and strategies that aim to induce or improve positive physical or psychological health outcomes" in nursing home residents [76]. In other words, interventions that: (1) incorporate technology (e.g., mobile

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applications), (2) utilize health promotional contents or strategies (e.g., information or emotional appeals), and (3) aim to improve nursing home residents' health outcomes (e.g., mental health status) will be included in the review.

It is important to note that, in this study, technology-based interventions are considered conceptually different from phenomena such as digital solutions. Compared to concepts such as digital solutions, technology-based interventions are unique as they are: (1) evidence-based: one of the main objectives of this study is to investigate and identify interventions that could offer the most benefits to nursing home residents based on empirical evidence, (2) purposeful: technology-based interventions are designed, developed, and/or delivered often tailored to the unique attributes of the target audience (e.g., nursing home residents), and (3) broader in scale and scope: technology-based interventions include interventions that are developed based on non-digital technologies (those that based on systems that generate and process binary data like 0 and 1), like landline telephones and audiotapes, which adopts analog technology (stores and interprets information in line with the devices physical properties) [77].

Overall, articles will be excluded if (1) the study sample did not include a majority of nursing home residents (i.e., nursing home residents ≤ 50% of the total research population), (2) the study did not focus on technology-based interventions (e.g., conventional interventions like face-to-face therapies), (3) study did not provide detailed information on the characteristics and effects of technology-based interventions (e.g., little or no information on the design or outcomes of the interventions), (4) the study did not report original and empirical findings on intervention outcomes (e.g., research protocols and systematic review studies), and (5) the study's main research method was not randomized controlled trials, experimental design, pre-post, quasi-experimental design, or observational studies.

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---Insert Table 1 here---

Search strategy

Databases including PubMed, PsycINFO, CINAHL, and Scopus, will be searched for eligible articles. We will also search ProQuest Dissertations to examine gray literature sources. A search strategy was developed in consultation with a librarian experienced in systematic review methods. Search terms used to locate articles will center on three concepts: nursing home residents, technology-based interventions, and randomized controlled trials. An example PubMed search string is illustrated in Table 2. Similar search terms will be adopted that are in accordance with: (1) the scale and scope of the databases and (2) insights and evidence accumulated in the review process.

---Insert Table 2 here---

Study selection

Following the search, all citations will be collated and uploaded to Mendeley, and duplicate studies will be removed. Titles and then abstracts will be screened by two principal reviewers independently. The same screening procedure will be adopted in the full-text article review process on selected article abstracts. Reasons for exclusion will be recorded and detailed in the PRISMA flowchart. Discrepancies between reviewers will be resolved via group discussions using videoconferencing and email correspondence to reach a consensus.

Study quality assessment

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The Cochrane Collaboration evaluation framework will be adopted to examine the risk of bias of the included study [78]. The framework has seven domains: random sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data, selective reporting, and any other source of bias. The risk of bias will be evaluated independently by two reviewers (ZS and XL), who will qualitatively evaluate the risk of bias and provide a score (high, medium, low). Any discrepancy regarding the risk of bias will be resolved by consensus via group discussions. Both junior and senior authors will be involved in the discrepancy review process, wherein, if agreement cannot be reached, the senior author will decide on the appropriate rating.

Data extraction and synthesis

Data on study design, sample characteristics (i.e., sample size and sample details), intervention characteristics (i.e., technology type, intervention application, intervention exposure, and intervention materials), outcome variables assessed, and research findings will be extracted by two main reviewers (ZS and XL). Findings from the included studies will be narratively synthesized to examine the characteristics and effects of the interventions. Gaining a more structured understanding of the interventions, the multidisciplinary study team will organize insights on intervention application and outcomes in tandem with the distinctive traits of nursing home residents and the overall nursing home environment. Due to the heterogeneity found within the articles identified during a preliminary review of search results, meta-analyses are not considered.

Ethics and dissemination:

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As the study is a protocol for a systematic review, ethical approval is not required. Ethical research practices, such as the guideline proposed by the United States National Institutes of Health [79], will be observed and abided by throughout the research continuum. The study findings will be disseminated via peer-reviewed publications and conference presentations.

Patient and Public Involvement

None.

Discussion

There is a growing body of technology-based interventions designed to support the health and quality of life of nursing home residents [52-58]. The onset of COVID-19 and recommended social distancing policy led to an increased interest in reliance on technology-based solutions [80, 81]. However, research has yet to provide comparative insight into the recent state of development of these interventions and how current evidence can be applied to the context of COVID-19. The use of the socio-ecological model, combined with multidisciplinary expertise, provides a framework to present practical insights on how these interventions can be utilized to deliver health solutions to nursing home residents amid and beyond the impact of COVID-19.

This research fills a critical gap in the literature by consolidating, in one place, the evidence for technology-based interventions empirically tested with nursing home populations. As the older adult population grows, there is an urgent need to identify effective technology-based interventions that can address the distinctive characteristics and preferences of nursing home residents [82, 83]. Improving person-centered care and the delivery of effective care solutions to nursing home residents, especially as the pandemic continues, is of critical

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importance. A comprehensive understanding of how available technology-based health solutions facilitate healthcare for nursing home residents can help shed light on approaches that are available to these residents to fend off the negative health consequences amid and beyond the influence of COVID-19. While the COVID-19 pandemic has revealed troubling vulnerabilities in the long-term care system across the globe, it also shows how telemedicine can support nursing home residents and their families. Technology can also assist clinicians in connecting with patients when in-person medical visits are difficult or dangerous (e.g., in rural settings, following natural disasters). Telemedicine and other technology-based interventions have the potential to provide a comprehensive range of benefits. This research also serves as a platform for care institutions and policymakers to inform government policies and further justify the role technology can play in strengthening the service provision across nursing homes and facilities.

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List of abbreviation

COVID-19: covronavirus disease 2019

Declarations

- Ethics approval and consent to participate
  - Not applicable.
- Consent for publication
  - Not applicable.
- Availability of data and materials
  - Data are available upon reasonable request.
- Competing interests
  - None.
- Funding
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  - ZS developed the research idea and wrote and manuscript. KM, YL, DMD, NMJ, XL, YD, SA, AC, JA, CPV, RYNC, JW, and HD reviewed and revised the manuscript.
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Table 1. Study inclusion criteria

Category	Criteria
Study population	Adults (≥18 years old) living in nursing homes
Language	English
Intervention/health solution	Technology-based interventions (e.g., educational materials delivered via smartphones and tablets, sensor devices, internet-based programs, etc.)
Key variable	Detailed descriptions of the technology-based interventions (i.e., purpose of the intervention (to evaluate the aims of existing interventions), use of technology (to examine how different types of technologies have been applied among nursing home residents), application of the interventions (to investigate the degree to which the interventions have involved nursing home residents in the adoption/application of technologies), intervention exposure (to study how different interventions have been used among nursing home residents), outcome variables assessed/measured (to evaluate the outcomes of technology-based interventions, such as improved physical or psychological health or wellbeing; more satisfactory patient-provider communication), and weather the design of the intervention material is tailored to nursing home residents (to examine to degree to which the design technology-based interventions has taken the unique attributes of nursing home residents into consideration)
Study type	Empirical studies
Study design	Randomized controlled trials, experimental design, pre-post, quasi-experimental design, and observational studies
Study outcome	Empirical reporting of the effect of the intervention (i.e., qualitative designs excluded)

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Table 2. Example PubMed search string

Concept	Search string
Nursing homes	"nursing home"[MeSH] OR "nursing home"[TIAB] OR "nursing homes"[MeSH] OR "nursing homes"[TIAB] OR "residential home*"[MeSH] OR "residential home"[TIAB] OR "caring home*"[TIAB] OR "home for the aged" [MeSH] "home for the aged" [TIAB] OR "long term care"[MeSH] OR "long term care"[TIAB] OR "senior housing"[TIAB] OR "assisted living facilities"[MeSH] OR "assisted living facilities"[TIAB] OR (senior[TIAB] OR geriatric[TIAB] OR elderly[TIAB] OR aged[TIAB] OR elder[TIAB] "older adults"[TIAB]) AND (housing[TIAB] OR living[TIAB])
Technology-based Interventions	"technology"[MeSH] OR "technology"[TIAB] OR "eHealth"[TIAB] OR "telemedicine"[MeSH] OR "telemedicine"[TIAB] OR "tele-medicine"[MeSH] OR "tele-medicine"[TIAB] OR "telehealth"[TIAB] OR "tele-health"[TIAB] OR "connected health"[TIAB] OR "digital health"[TIAB] OR "mHealth"[TIAB] OR "mobile health"[TIAB]
Randomized controlled trials	randomized controlled trial[PT] OR randomized controlled trials as topic[MH] OR random allocation [MH] OR double-blind method[MH] OR single-blind method[MH] OR random*[tw] OR "Placebos"[MeSH] OR placebo[TIAB] OR ((singl*[tw] OR doubl*[tw] OR trebl*[TW] OR tripl*[TW]) AND (mask*[TW] OR blind*[TW] OR dumm*[TW]))
COVID-19	((coronavirus OR "corona virus" OR coronavirinae OR coronaviridae OR betacoronavirus OR covid19 OR "covid 19" OR nCoV OR "CoV 2" OR CoV2 OR sarscov2 OR 2019nCoV OR "novel CoV" OR "wuhan virus") OR ((wuhan OR hubei OR huanan) AND ("severe acute respiratory" OR pneumonia) AND (outbreak)) OR "Coronavirus"[Mesh] OR "Coronavirus Infections"[Mesh] OR "COVID-19" [Supplementary Concept] OR "severe acute respiratory syndrome coronavirus 2" [Supplementary Concept] OR "Betacoronavirus"[Mesh])

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Figure 1. An ecological model of factors increasing nursing home resident vulnerability to COVID-19

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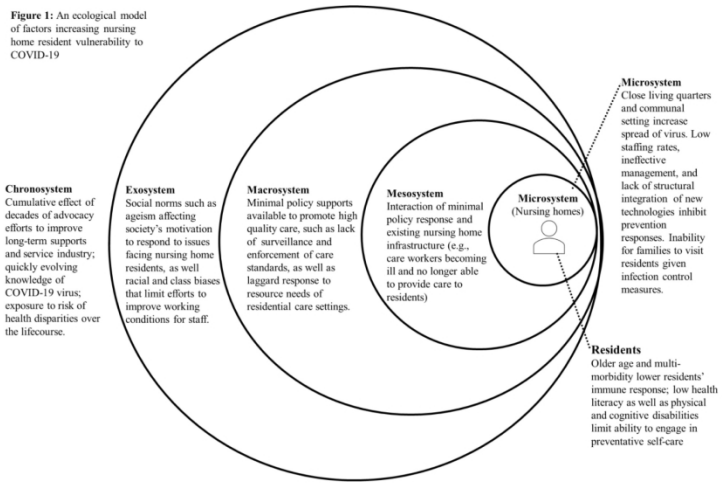


Figure 1. An ecological model of factors increasing nursing home resident vulnerability to COVID-19

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Page
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	2
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	NA
Support:			
Sources	5a	Indicate sources of financial or other support for the review	14
Sponsor	5b	Provide name for the review funder and/or sponsor	
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	4-7
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4-7
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	8-11
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	8-9
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	10
Study records:			

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Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	10-11
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	10-11
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	10-11
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	10-11
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	20
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	10-11
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	11
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	11
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	10-11
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	10-11
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	11
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	10-11

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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